

HARSH RAJ

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Google Scholar ◇ Github ◇ LinkedIn

EDUCATION

Northeastern University

M.S. in Computer Science

GPA: 3.4/4.0

Related courses: Foundations of AI, Algorithms

May 2026 (expected)

Delhi Technological University (DTU)

B.Tech. in Engineering Physics

GPA: 3.4/4.0

Related courses: Machine Learning, Algorithmic Game Theory, Computer Vision.

June 2023

RESEARCH INTERESTS

I am interested in large-scale data processing systems and training large models at scale, which have been my current focus as a co-founder of OntocordAI.

EXPERIENCE

Co-Founder (CTO), Ontocord AI

Remote

Oct 2024 – Present

California, US

- Training AuroraM-2 on the Leonardo supercomputer. Optimized the training infrastructure for low-communication, multi-node supercomputing environments.
- Developed the multimodal data synthesis pipeline.

Research Engineer, AI Risk and Vulnerability Alliance (ARVA)

Remote

Oct 2024 – Present

Washington, US

- Contributed multiple bias benchmarks to the UK AISI inspect-ai framework.
- Integrated the inspect-ai framework with the AVID database.

Applied Scientist, Vijil AI

Remote

Feb 2024 – Oct 2024

California, US

- Developed vijil-fuzzer, an LLM red-teaming framework for enterprise-grade customers.
- Integrated open-source benchmarks into the vijil-evaluation service.
- Contributed to vijil-dome, integrating guardrails into tools like Langchain with minimal latency impact.

ML Engineer, Yield Protocol

Remote

Sept 2022 – Jan 2024

Chicago, US

- Led data synthesis to improve LLM agents and built synchaev, a framework for agent data generation.
- Developed mandrill for finetuning and testing agents. Supervised 10 research interns at UIUC's Disruption Lab as part of the project.
- Contributed to the development of Cacti, a Web3 transaction chatbot.

Applied Researcher (intern), Thoucentric

Remote

May 2022 – Aug 2022

Bangalore, India

- Developed a natural language to SQL conversion system for data analytics including automatic visualization.

- Improved the previous state-of-the-art by introducing additional processing steps, boosting accuracy by 10%. Authored a white paper.

Data Scientist (intern), Attryb Tech

On-Site

June 2021 – Dec 2021

Bangalore, India

- Developed the backend for Content Studio, a tool for generating and analyzing content for blogs, product marketing, etc.
- Finetuned Pegasus and GPT-2 for multiple components of the content generation pipeline.
- Mined data from SemRush and Common Crawl for content database creation.

PUBLICATIONS

- **Harsh Raj**, Vipul Gupta, Domenic Rosati, Subhabrata Majumdar. *Improving Consistency in Large Language Models through Chain of Guidance* (TMLR'2025).
- Domenic Rosati, Giles Edkins, **Harsh Raj**, David Atanasov, Kai Williams, Subhabrata Majumdar, Janarthanan Rajendran, Frank Rudzicz, Hassan Sajjad. *Defences against Reverse Preference Attacks* (AI Security Workshop, AAAI'25).
- Laura Fee Nern, **Harsh Raj**, Maurice Georgi, Yash Sharma. *On Transfer of Adversarial Robustness from Pretraining to Downstream Tasks* (NeurIPS'23).
- **Harsh Raj**, Domenic Rosati, Subhabrata Majumdar. *Measuring Reliability of Large Language Models through Semantic Consistency* (Best Paper Award, ML Safety Workshop, NeurIPS'22).
- Sinchani Chakraborty, **Harsh Raj**, Srishti Gureja, Tanmay Jain, Atif Hassan, Sayantan Basu. *Evaluating the Robustness of Biomedical Concept Normalization* (Transfer Learning Workshop, NeurIPS'22).
- **Harsh Raj**, Ashutosh Pandey, Shaurya Kumar, Kavinder Singh, Nihal Kumar, Anil Singh Parihar. *Decoding Percepts in Vision Language Navigation: Is it about better features or more data?* (ACM TIST).
- Manu Joseph, **Harsh Raj**. *GANDALF: Gated Adaptive Network for Deep Automated Learning of Features* (under review, TMLR).
- Manu Joseph, **Harsh Raj**, Abhinav Yadav, Aaryamann Sharma. *AskYourDB: An end-to-end system for querying and visualizing relational databases using natural language*.
- **Harsh Raj**, Aakansha Gupta, Rahul Katarya. *Extract It! Product Category Extraction by Transfer Learning* (CICT'22).

ACHIEVEMENTS

Vision and Language Navigation: Secured 3rd position on the R2R benchmark	<i>2022</i>
ML Safety Challenge (NeurIPS'22): Best Paper Award with a \$5000 prize	<i>2022</i>
Codeforces ML Round: Rank 7 in the Raif ML Round 1	<i>2021</i>

SKILLS/HOBBIES

Programming Languages	Python, C/C++, MATLAB, HTML
Machine Learning Tools	Pytorch, Tensorflow, Sklearn, Pandas, Numpy
Hobbies	birding and hiking